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[23CE32]

THIRD SEMESTER B. Tech EXAMINATION, DECEMBER-2024

SURVEYING - I

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	5 X 2=10		
			BL	COs	Marks
1.	a.	What is geodetic surveying?	2	1	02
	b.	What are the three primary techniques used in plane table surveying for mapping and data collection?	2	2	02
	c.	What are three commonly used instruments for determining levels in surveying?	2	1	02
	d.	How can the area of an irregular shape be calculated using average ordinate method?	2	2	02
	e.	What is contour interval, and how is it defined in the process of contouring?	2	1	02
		PART-B	**6 X 5=30**		
2.	a.	What is chain surveying, and how is it used for measuring distances and mapping small areas using linear measurements?	2	1	05
		OR			
2.	b.	Write a brief note on the different instruments required for linear measurement	2	2	05
3.	a.	List the different temporary adjustments of Plane Table Surveying	2	3	05
		OR			
3.	b.	Explain the whole circle and quadrant circle bearing systems with a neat sketch to illustrate the differences between them	2	2	05
4.	a.	The following consecutive readings were taken with a level and 5M leveling staff on continuously sloping ground at a common interval of 20M. 0.385, 1.030, 1.925, 2.825, 3.730, 4.685, 0.625, 2.005, 3.110, 4.485. The Reduced Level of the first point was 208.125M. Rule out a page of a level field book and enter the above readings. Calculate the reduced levels of the points by Rise and Fall Method and also the Gradient of the line joining the first and the last point.	3	3	05

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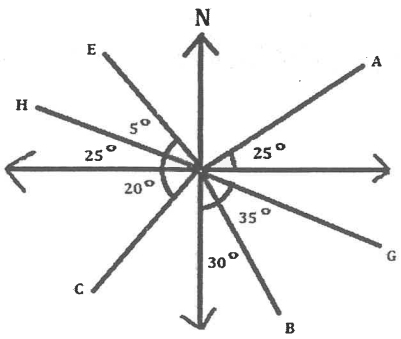
		OR			
4.	**b.**	Explain the various types of leveling in surveying.	2	2	**05**
5.	**a.**	The following perpendicular offsets were taken at 10 meters intervals from a survey line to an irregular boundary line. 3.25, 5.60, 4.20, 6.65, 8.75, 6.20, 3.25, 4.20, 5.65 Calculate the areas enclosed between all the lines using Average Ordinate Rule.	3	4	**05**
		OR			
5.	**b.**	The following perpendicular offsets were taken at 10 meters' intervals from a survey line to an irregular boundary line. 3.25, 5.60, 4.20, 6.65, 8.75, 6.20, 3.25, 4.20, 5.65 Calculate the areas enclosed between all the lines using Trapezoidal Rule.	3	4	**05**
6.	**a.**	What are the key characteristics of contour lines in surveying	2	2	**05**
		OR			
6.	**b.**	What are the various methods of interpolation used in contouring?	2	2	**05**
7.	**a.**	Enumerate and elucidate minor tools for establishing right angles.	2	2	**05**
		OR			
7.	**b.**	What are the different types of compasses used in surveying, and how do they differ in terms of design, functionality, and application?	3	3	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	What are the fundamental principles of surveying?	2	2	**10**
		OR			
8.	**b.**	What are the various classifications of surveying based on the instruments used, and explain each category?	3	3	**10**
9.	**a.**	Convert the angles in to whole circle bearing and quadrant bearing for the given figure. 9.  	4	5	**10**

		Figure. 9			
		OR			
9.	**b.**	Illustrate procedure for acquiring ground data utilizing Intersection methods, including implementing temporary adjustments	4	5	**10**
10.	**a.**	A page of a level book was defaced so that the only legible figures were Consecutive entries in the column of reduced level, 55.565(B.M), 54.985(TP), 55.170, 56.265, 53.670, 53.940(TP), 52.180, 52.015, 51.480, (TP), 53.145, 54.065(TBM). Entries in Back Sight Column, 1.545, 2.310, 0.105, 3.360 in order from top of the page. Reconstruct the page as booked and check your work.	4	5	**10**
		OR			
10.	**b.**	The following values were extracted from a level field book. Some of the entries being illegible owing to exposure to rain. Insert the missing values and check your results. Re-book all the figures by the Rise and Fall Method.	4	5	**10**

Station	BS	IS	FS	RISE	FALL	R. L	Remark
A	2.285					232.460	BM 1
B	1.650		X	0.020			
		2.105			X		
C	X		1.960	X			
D	2.050		1.925		0.300		
		X		X		232.255	BM 2
E	1.690		X	0.340			
F	2.865		2.100		X		
			X	X		233.425	BM 3

11.	**a.**	The following perpendicular offsets were taken from a chain line to a hedge Calculate the area between all the lines by using Simpson's Rule.	4	5	**10**

CHAINAGE (M)	OFFSETS (M)
0	7.6
15	8.5
30	10.7
45	12.8
60	10.6
70	9.5
80	8.3
100	7.9
120	6.4
140	4.4

		OR			
11.	**b.**	The Following perpendicular offsets were taken from a chain line to an irregular boundary. Calculate the area between the chain line, the boundary and the end offsets by trapezoidal rule.	4	5	**10**

CHAINAGE (M)	0	10	25	42	60	75
OFFSET (M)	15.5	26.2	31.8	25.6	29.0	31.5

12.	**a.**	What is an L-section in contouring, and how is it used to represent the elevation profile of a land surface along a specific line?	3	3	**10**
		OR			
12.	**b.**	What are the indirect methods of contouring, and how do they differ from direct methods in representing landforms and elevations?	2	2	**10**
13.	**a.**	The following perpendicular offsets were taken from a chain line to a hedge CHAINAGE (M): 0, 15, 30, 45, 60, 70, 80, 100, 120, 140. OFFSETS (M): 7.6, 8.5, 10.7, 12.8, 10.6, 9.5, 8.3, 7.9, 6.4 and 4.4. Calculate the area between all the lines by using Trapezoidal Rule.	4	5	**10**
		OR			
13.	**b.**	What are the various methods for conducting theodolite traverses, and can you explain each method in detail?	3	3	**10**